

AI-Powered IoT-Based Microclimate Control System for Smart Indoor Farming

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Introduction

Food security and sustainable agriculture have become pressing global challenges due to rapid urbanization, climate change, and limited arable land. Indoor farming and greenhouse cultivation are emerging as innovative solutions, enabling controlled environments for crop production. However, manually managing microclimatic parameters such as temperature, humidity, soil moisture, and light intensity can be labor-intensive and inconsistent, especially when scaled.

The advancement of Artificial Intelligence (AI) and Internet of Things (IoT) provides a unique opportunity to address these challenges. IoT sensors can continuously capture environmental parameters, while AI algorithms can make real-time decisions to optimize crop growth. In particular, Reinforcement Learning (RL) allows the system to learn and adapt from user interactions and environmental feedback, providing intelligent automation. To ensure trust, transparency, and adoption, Explainable AI (XAI) will be integrated so that users understand why the system takes specific actions.

This project aims to design and prototype a smart AI-powered microclimate control system that autonomously manages indoor crop environments. For the current stage, a scaled-down version has been implemented in a controlled “indoor farming box” using IoT sensors and actuators with predefined thresholds.

Work Done and Proposed Model

1. Current Prototype

In the preliminary implementation, a small-scale enclosed setup (the box model) was designed to replicate an indoor farming environment. A plant already thriving in the existing weather condition was placed inside this box to simulate realistic growth requirements.

Sensors used:

- ❖ DHT11 → Temperature & Humidity sensing

- ❖ LDR/Light Sensor → Light intensity measurement
- ❖ Soil Moisture Sensor → Root-zone water availability

Actuators used:

- ❖ Ventilation Fan → Controls air circulation and temperature
- ❖ Artificial Light → Provides supplemental lighting
- ❖ Water Pump → Supplies water when soil dries

The sensors record real-time environmental data. Each actuator is programmed to work at a pre-set threshold (derived from current environmental conditions suitable for the chosen plant).

For example:

If temperature > threshold → Fan ON

If soil moisture < threshold → Water pump ON

If light < threshold → Artificial light ON

This setup ensures that the plant continues to thrive in an artificially maintained indoor environment.

2. Proposed Advanced Model with AI Integration

- a. While the threshold-based model provides proof of concept, the long-term vision involves integrating Reinforcement Learning (RL)
- b. Learning from user interactions: Initially, the user operates the actuators manually for a few days, depending on plant needs. Sensors record these decisions alongside environmental conditions.
- c. Policy development: The RL agent identifies patterns in these user interventions, gradually learning optimal control strategies.
- d. Autonomous adaptation: Once trained, the system autonomously regulates actuators in real-time, adjusting continuously to optimize growth.
- e. Explainability (XAI): Every action taken by the AI will be accompanied by an explanation (e.g., “Fan switched ON because humidity exceeded safe range for crop growth”), ensuring user trust.

This design ensures that the system does not merely follow static thresholds but evolves into a self-learning, adaptive, and farmer-friendly solution.

Results (Preliminary Stage)

- a. Sensor-Actuator Response: The threshold-based box setup successfully demonstrated environmental monitoring and actuation. Environmental parameters inside the box stabilized within the desired range, confirming feasibility.
- b. Plant Health: The selected plant maintained steady growth conditions indoors, validating that the system can replicate favourable conditions.
- c. User Interaction: Early tests indicate that the system can react quickly to environmental deviations, reducing manual intervention.

While these results are preliminary, they form a strong foundation for transitioning into AI-based adaptive control.

Conclusion

The project demonstrates the feasibility of creating a smart indoor farming system that integrates IoT sensing with AI-based control strategies. The current threshold-based prototype ensures environmental stability, while the proposed RL-based extension will enable intelligent, adaptive learning from user interactions. With XAI integration, the system also emphasizes transparency, making it both technically powerful and user-friendly.

This work has the potential to evolve into a scalable product that supports sustainable agriculture, optimizes energy and water usage, and empowers farmers and urban growers with advanced AI tools.

References

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